

Topics:

- ☐ Approximating definite integrals (Area) by summing the areas of rectangles.
- ☐ Rewrite and/or compute an integral using the properties of integrals.
- ☐ Compute a definite integral of f over an interval by plugging the endpoints of the interval into the anti-derivative of f (Fundamental Theorem of Calculus Part 2).
- ☐ Compute an indefinite integral (i.e. anti-derivative).
- ☐ Find an integral using the substitution technique.
- ☐ Take the derivative of an integral when one of the endpoints is a variable (Fundamental Theorem of Calculus Part 1).

Items of discussion. In addition to your questions, we will also go over:

- ☐ Exam 1, Problem 2 (Approximating the derivative)
- ☐ Exam 2, Problem 10 (Intrepreting the derivative)

Practice Problems:

1. Compute the following indefinite integrals:

(a) $\int (x^5 - x) dx$

(e) $\int \frac{3}{x^5} dx$

(b) $\int x \left(x^2 + \frac{1}{x} \right) dx$

(f) $\int \frac{1}{1 + 2x} dx$

(c) $\int x \sqrt{x^2 - 1} dx$

(g) $\int 3x^2 e^{x^3} dx$

(d) $\int \frac{2x - 1}{(x^2 - x)^2} dx$

(h) $\int x^{1/3} dx$

2. Compute the following definite integrals:

(a) $\int_1^4 (8x^3 - x) dx$

(c) $\int_{-3}^{-1} 7 dx$

(b) $\int_1^2 \left(e^x + \frac{1}{x} \right) dx$

(d) $\int_0^1 x(x^2 + 7)^6 dx$

3. Let $F(q) = \int_3^q \sqrt{t^2 - 1} dt$. Compute $F'(q)$ using the Fundamental Theorem of Calculus part 1.

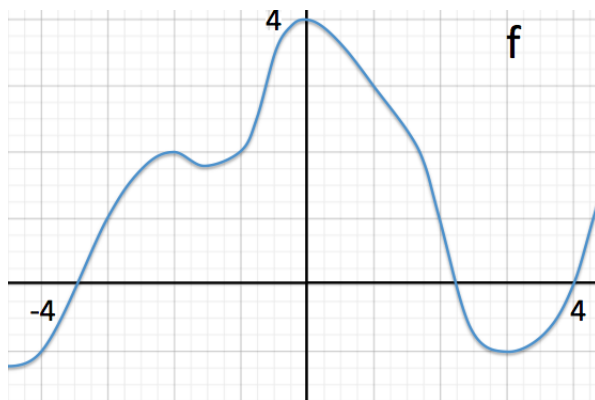
4. If $\int_9^{26} f(x) dx - \int_9^{25} f(x) dx = \int_a^b f(x) dx$, then $a =$, and $b =$

5. Suppose MAJ Libertini is driving to conference in a car that has a broken odometer (the meter that tells you the mileage of a car). In order to get a reimbursement on travel costs, she must report the mileage (assume she doesn't know about google maps). Fortunately, her speedometer still works and she can use this fact to get a decent approximation of the distance travelled. The total trip took an hour and table lists the readings she took of her speedometer every 15 minutes.

time (min)	Speed (mph)
0	0
15	60
30	42
45	78
60	0

- (a) Based on the data, what is the smallest Δt we can use in our Riemann Sum?
- (b) Sketch two identical plots of the data side by side.
- (c) On one plot from (b), draw rectangles (with left-hand heights) and approximate the total distance MAJ Libertini drove.
- (d) On the other plot from (b), draw rectangles (with right-hand heights) and approximate the total distance MAJ Libertini drove.
- (e) What could have MAJ Libertini done differently to improve her approximation.

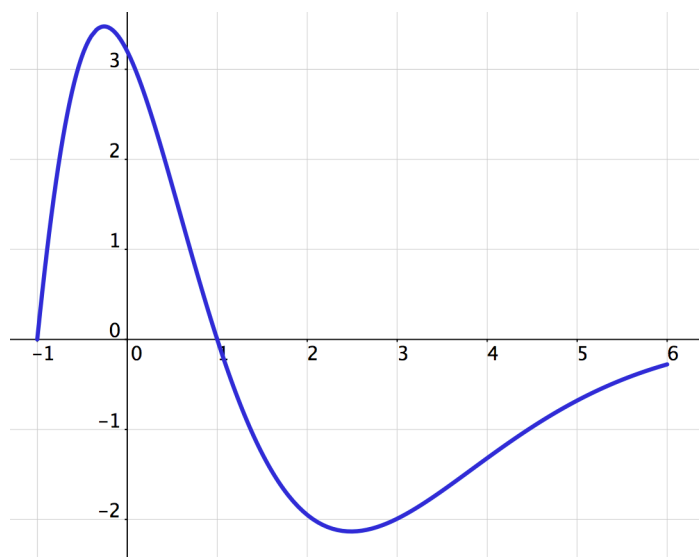
6. The graph of f is given.



- (a) Using the right-height rule & 4 rectangles, approximate $\int_{-4}^4 f(x) dx$
- (b) Using the left-height rule & 4 rectangles, approximate $\int_{-4}^4 f(x) dx$

7. If $\int_0^1 3f(x) dx = 15$, then $\int_0^1 f(x) dx =$

8. Below is the graph of some function, $g(x)$ on the interval $[-1, 6]$. Using the graph, answer the questions that follow.



(a) Fill in each box with either $<$, $>$, or $=$.

- | | | |
|--|---|--|
| • $\int_{-1}^0 g(x) dx \quad \square \quad 0$ | • $\int_{-1}^2 g(x) dx \quad \square \quad 0$ | • $\int_1^6 g(x) dx \quad \square \quad -2$ |
| • $\int_{-1}^1 g(x) dx \quad \square \quad 8$ | • $\int_0^1 g(x) dx \quad \square \quad \int_1^0 g(x) dx$ | • $\int_1^6 g(x) dx \quad \square \quad \int_0^6 g(x) dx - \int_0^1 g(x) dx$ |
| • $\int_1^4 g(x) dx \quad \square \quad g'(4)$ | • $\int_{-1}^1 g(x) dx \quad \square \quad \int_{-1}^2 g(x) dx$ | • $\int_{-1}^6 g(x) dx \quad \square \quad \int_{-1}^1 g(x) dx$ |

9. Answer the following true/false questions:

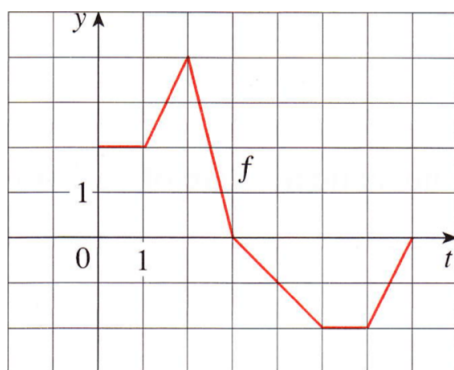
(a) (TRUE FALSE) $\int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx$

(b) (TRUE FALSE) $\int_a^b f(x) dx + \int_a^b f(x) dx = 2 \int_a^b f(x) dx$

(c) (TRUE FALSE) $\int_a^b (x^2 - 1)(x + 1)^4 dx = \left(\int_a^b x^2 - 1 dx \right) \left(\int_a^b (x + 1)^4 dx \right)$

10. If $\int_6^{12} f(x) dx = 6$, $\int_6^8 f(x) dx = 3$, $\int_{10}^{12} f(x) dx = -2$, then $\int_8^{10} f(x) dx =$

11. The graph of f and some information about $g(x)$ are given.



$$\int_0^7 g(x) dx = -5$$

$$\int_1^7 g(x) dx = 4$$

$$\int_0^3 g(x) dx = 6$$

Using the above information about f and g , compute the following:

(a) $\int_0^1 f(x) dx =$

(b) $\int_0^7 f(x) dx =$

(c) $\int_0^1 g(x) dx =$

(d) $\int_0^1 (f(x) - g(x)) dx =$

(e) $\int_0^7 (3f(x) + 2g(x)) dx =$

(f) $\int_1^3 4g(x) dx =$